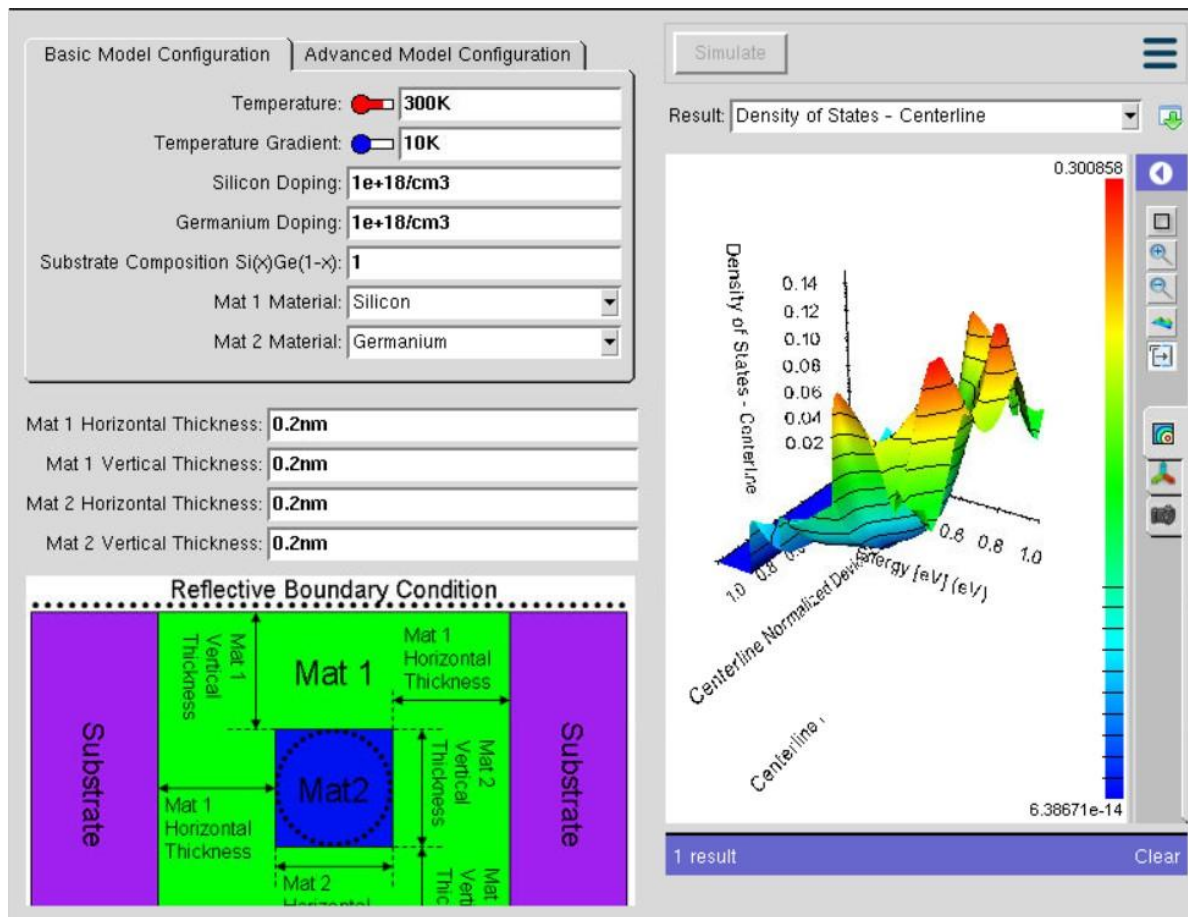


Exp-2

Exp. 2. Simulation of Density of States of Quantum Well Structure with Various Nanometer Thickness.

Aim: To simulate density of states of quantum well structure with various nano meter thickness.

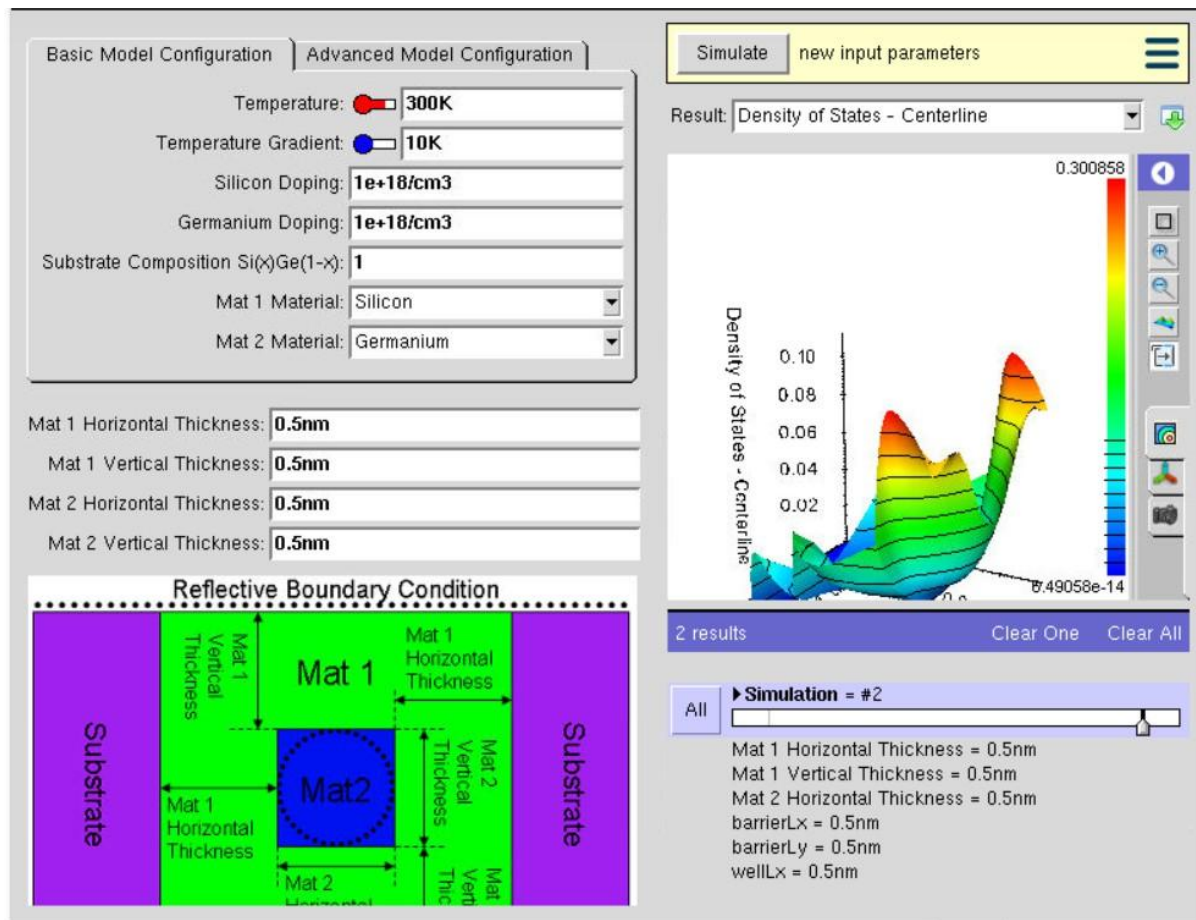


This graph shows the **Density of States (DOS) along the centerline** of the Silicon–Germanium (Si–Ge) nanostructure.

- It illustrates how the **available electron energy states** vary with **energy and position** inside the device.
- The **peaks (red/yellow regions)** represent a **high density of energy states**, where electrons are more likely to exist.
- The **low regions (blue/green)** indicate **fewer available energy states** for electrons.
- This graph helps analyze the **electronic behavior, carrier distribution, and transport properties** of the Si–Ge nanostructure, which are important for device performance.

Exp-2 case study-1

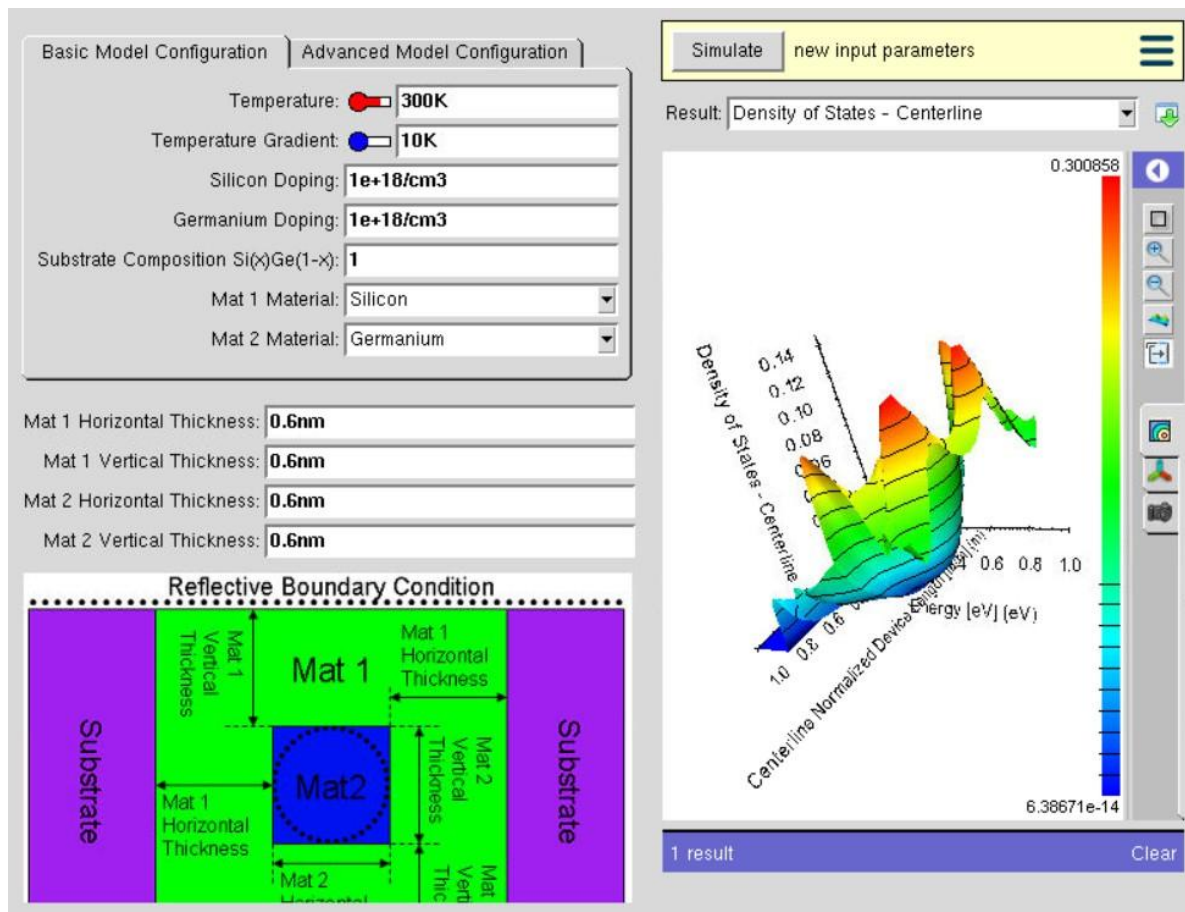
Case Study 1: Simulate the density of states of quantum well of thickness 0.5 nm



This graph also represents the **Density of States (DOS) along the centerline** for the Si-Ge nanostructure, but with the **thickness increased to 0.5 nm**.

- The graph shows how the **available electron energy states** are distributed in the nanostructure at **0.5 nm thickness**.
- The **red and yellow peaks** indicate regions with a **high density of electron states**, while the **blue and green regions** indicate lower density.
- Compared to the **0.2 nm structure**, the DOS distribution becomes **smoother and broader**, indicating changes in electron confinement due to increased thickness.
- This graph is useful for evaluating how **thickness affects the electronic properties and carrier transport** of the Si-Ge nanostructure.

Case Study 2: Simulate the density of states of quantum well of thickness 0.6 nm



This graph represents the **Density of States (DOS) along the centerline** of the Si-Ge nanostructure with the **thickness increased to 0.6 nm**.

- The graph shows the **distribution of available electron energy states** for the **0.6 nm thick Si-Ge structure**.
- The **high peaks (red/yellow regions)** indicate a **higher density of electron states**, while the **blue/green regions** represent lower density.
- Compared to **0.2 nm** and **0.5 nm**, the **0.6 nm thickness produces broader and more continuous DOS peaks**, indicating reduced quantum confinement and improved carrier transport.
- This result helps analyze how **increasing the material thickness influences the electronic properties and overall device performance**.

In conclusion, The simulation results show that the **Density of States (DOS)** of the Silicon–Germanium (Si–Ge) nanostructure changes as the material thickness increases from **0.2 nm to 0.6 nm**. At **0.2 nm**, the DOS exhibits sharp and narrow peaks due to strong quantum confinement of electrons. As the thickness increases to **0.5 nm**, the peaks become broader and smoother, indicating reduced confinement and improved electron mobility. At **0.6 nm**, the DOS becomes more continuous with wider energy distributions, allowing better carrier transport through the device. Overall, increasing the Si–Ge layer thickness enhances the electronic properties by reducing quantum confinement effects and improving the transport characteristics, which can lead to better device performance.